Use of Deep Learning in Healthcare Systems for Network Attack Detection and Mitigation: A Literature Review (2026)

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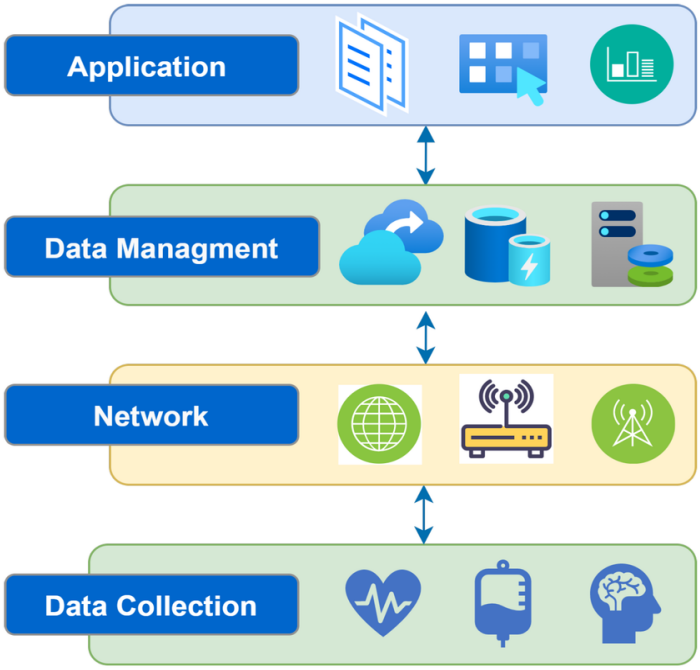
***Abstract*—Healthcare systems increasingly rely on Interconnected medical devices and internet of Medical Things (ioMT) infrastructure to deliver patient care, Monitor vital signs and manage clinical operations. This connectivity exposes network to various cyber Threats, including denial-of-service-attacks, man-in-the-Intrusions, and malware infection targeting sensitive medical Data and device functionality. Deep Learning techniques Provide advanced capabilities for detecting anomalous Network behavior and mitigation such attack through pattern Recognition in high-dimensional traffic data. This literature Review examines recent and foundational studies on deep Learning applications in healthcare cybersecurity, including Convolutional neural networks, recurrent models, hybrid Reinforcement learning approaches, and federated learning For privacy preservation. The analysis reveals significant Advancements in real-time anomaly detection and intrusion Prevention, alongside persistent challenges such as limited Domain-specific datasets and scalability in edge environments. Implications for cybersecurity practice and directions for Future research are discussed**

***Index Terms*—Anomaly detection, intrusion detection, Deep learning, Healthcare systems, Internet of Medical Things, Cybersecurity, Network attacks, Reinforcement learning**

I. INTRODUCTION

HEALTHCARE systems increasingly rely on interconnected medical devices and internet of Medical Things (IoMT) infrastructure to deliver patient care, monitor vital signs, and manage clinical operations. This connectivity exposes networks to various cyber threats, including denial-of-service attacks, man-in-the-middle intrusions, and malware infections targeting sensitive medical data and device functionality [3],
[6]. Deep learning techniques provide advanced capabilities For detecting anomalous network behavior and mitigating such attacks through pattern recognitions in high dimensional Traffic data [9], [12].



Network attacks in healthcare environments affect millions of individuals annually. Data breaches compromise protected health information for hundreds of thousands of patients each year, Leading to delayed treatments, identity thefts, and Financial losses exceeding billions of dollars globally [3], [4]. The various stakeholders involved include patients whose privacy and safety are compromised, healthcare providers who experience disruptions in service delivery, healthcare administrators who have the responsibility of ensuring compliance, government agencies that have the responsibility of enforcing compliance with data protection regulations, and medical device manufacturers who have the responsibility of designing medical devices securely [3], [6].

The importance of this topic is due to the fact that the rate of IoMT usage is growing, which increases the size of the surfaces of vulnerability in critical infrastructures. The detection and mitigation of network attacks using deep learning can improve the resilience of a system, ensure patient safety, and promote regulatory compliance in a world with increasing cyber risks [2], [12].

II. RESEARCH STATEMENT/ OBJECTIVES

This paper reviews existing literature on the use of deep learning techniques for network attack detection and mitigation in healthcare networks, including those related to Internet of Medical Things (IoMT) technologies and medical applications. It also examines existing as well as seminal studies on deep learning technologies, including convolutional networks, recurrent networks, reinforcement learning, as well as federated learning technologies, to determine their overall impact on network cybersecurity.

The main goal is to provide a comprehensive synthesis of the current literature pertaining to deep learning-based intrusion detection systems (IDS) for healthcare networks. This requires assessing the effectiveness of various approaches in identifying and mitigating different types of attacks, ranging from common ones such as denial of service, man-in-the-middle attacks, and spoofing, to emerging ones such as malware affecting patient information and medical devices. The main focus of this review will be on key issues identified in the literature, particularly in regards to ensuring accurate detection in a high-dimensional environment, ensuring patient information privacy in accordance with healthcare regulations and laws, the requirement for real-time processing in accordance with IoT and IoMT devices' limitations, and the limitations of current approaches in this realm, which include issues of data imbalance and interpretability.

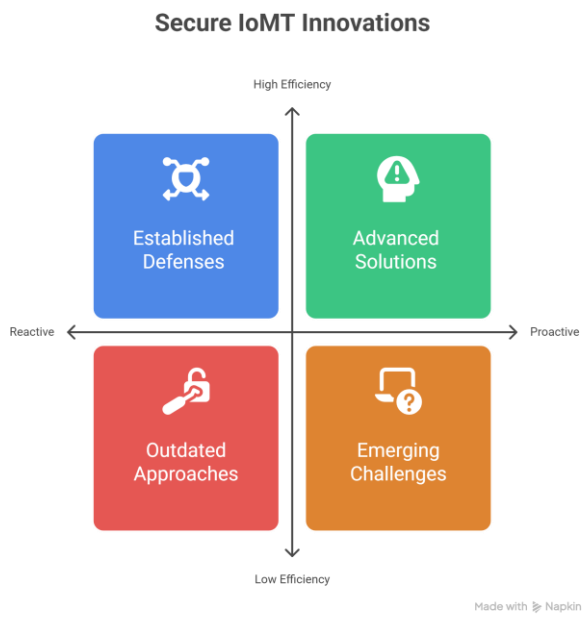
Additionally, this study aims to identify the degree of implementation of deep learning in addressing cybersecurity issues in the IoMT domain. It also aims to highlight new trends in addressing cybersecurity issues in the IoMT domain through the integration of federated learning in ensuring the privacy of patient information and the integration of reinforcement learning in ensuring the mitigation of threats in a 5G-enabled network. Through a systematic analysis of the strengths and limitations of each of these approaches, this review aims to highlight their implications in ensuring the enhancement of network resilience and in reducing false positive rates in a healthcare network.

The purpose of this systematic review of the literature is to identify the current knowledge gap and propose directions for future research that can lead us towards safe and intelligent healthcare systems.

III. REVIEW OF THE LITERATURE

The body of literature on the application of deep learning techniques in the detection and mitigation of network attacks in healthcare systems has experienced a significant increase in recent years. This increase is mainly driven by the extensive use of Internet of Medical Things devices and the sophisticated nature of network attacks targeting medical infrastructures. The current section builds on the introduction by incorporating both emerging and traditional literature on network attacks and mitigation strategies in healthcare systems. The incorporation of literature is done in a way that promotes clarity and a comprehensive understanding of the current state of cybersecurity literature, the nature of the activities, and their implications for future studies.

The rapid incorporation of IoMT devices in the healthcare environment has allowed for remote monitoring, real-time diagnosis, and personalized treatments, but at the same time, it has created a larger attack vector for cyber attackers. The initial studies in this field have historically focused on traditional machine learning techniques and rule-based intrusion detection systems. However, these have been found to have limitations in dealing with the changing and sheer volume of IoMT-based network traffic. Modern studies have shifted towards the application of deep learning techniques, which have better feature extraction and pattern detection capabilities, particularly in dealing with high-dimensional time-series data.



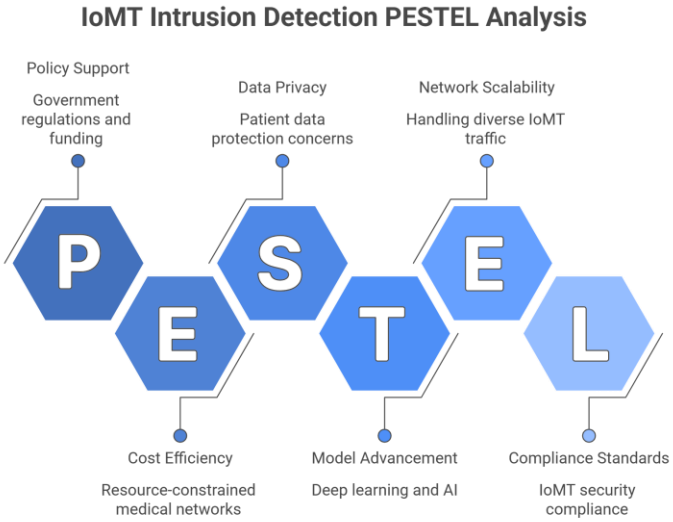
This change is a reflection of the overall trend in the field of cybersecurity research in the healthcare industry, which has seen a move from reactive signature-based systems to more proactive anomaly-based systems with the ability to protect against zero-day attacks. The sheer volume of publications on the topic since 2020, with over 400 studies focusing on the application of deep learning in the field of IoMT security, speaks to the immediacy of the need given the reliance on connected systems during and after the COVID-19 pandemic. As a whole, they provide a body of work on which the state of the field and the issues yet to be overcome can be evaluated.

IV. DEEP LEARNING TECHNIQUES FOR INTRUSION DETECTION

Deep learning models have been found to perform well for intrusion detection in IoMT networks, as they have been able to effectively deal with high-dimensional network data. Convolutional neural networks (CNNs), when combined with recurrent layers such as long short-term memory (LSTM), have been able to perform intrusion detection with high accuracy for malicious traffic patterns [9], [13]. These models have been able to effectively learn spatial features from packet data as well as temporal features for intrusion detection in resource-constrained medical networks [10].

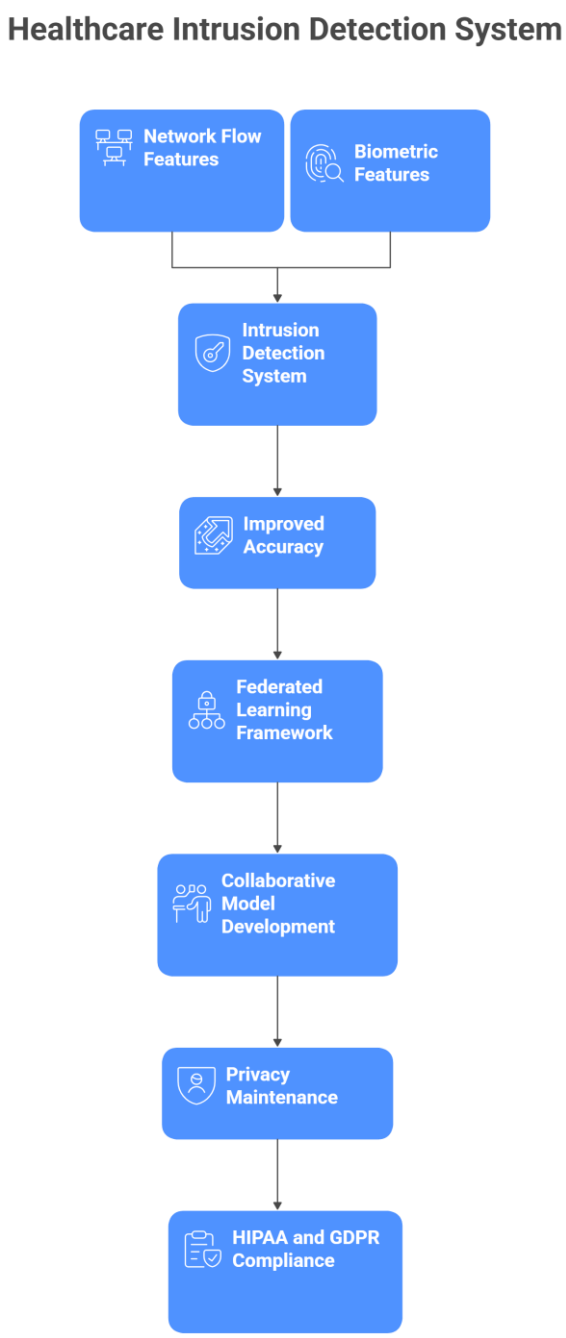
Hybrid reinforcement learning approaches provide additional robustness through the support of adaptive decision-making in the face of changing threats. This can be achieved through the integration of DDPG and convolutional-recurrent networks for the development of a framework that learns the best mitigation actions in a changing 5G medical environment [7], [11]. Feature engineering approaches, particularly those that use an attention mechanism and mean decrease in accuracy (MDA), provide efficiency in the model through dimensionality reduction and the use of the most relevant features [8].

Conventional studies have mostly relied on supervised learning for known attack signatures, whereas recent studies have started employing unsupervised and semi-supervised learning to deal with the problem of unlabeled data that is frequently encountered in healthcare environments [12], [15]. This suggests a trend towards more flexible and scalable solutions that can handle the diversity of IoMT traffic.



V. APPLICATIONS IN HEALTHCARE AND IOMT ENVIRONMENTS

In healthcare-specific applications, intrusion detection systems combine network flow features with biometric features. The integration of these two feature types has shown improvement in the accuracy of intrusion detection ranging from 7% to 25% as compared to using a single feature type [1]. The federated learning framework allows collaborative model development without sharing patient information between healthcare nodes, thus maintaining privacy as per various healthcare regulations such as HIPAA and GDPR [2].



Bluetooth-enabled vulnerabilities in smart healthcare devices are a major area of concern for researchers. The community has developed datasets and deep learning models that have achieved almost perfect accuracy in identifying man-in-the-middle and denial-of-service attacks in Bluetooth Low Energy-based communication systems [6]. The research activity in this area has been increasing exponentially since 2020, with over 400 publications related to the use of deep learning for IoMT security. This can be attributed to the increased reliance on remote healthcare systems during the current health crisis, including the COVID-19 pandemic [2], [12].

VI. MITIGATION STRATEGIES AND PERFORMANCE EVALUATION

In recent times, mitigation strategies are increasingly employing ensemble methods and hybrid models to effectively classify multi-class attacks. Ensemble tree-based classifiers apply the methods developed for industrial IoT networks to healthcare scenarios, which result in improved F1-scores for spoofing and scanning attack detection [5]. Hybrid models that combine deep learning with bidirectional encoder representations from transformers (BERT) improve attack detection for healthcare electronics by effectively processing sequential data, which results in improved accuracy by up to 3.9 percent compared to conventional models [14].

The performance evaluation using datasets such as CICIoMT2024 and ToN-IoT has revealed that it is possible to achieve 97 percent accuracy using LSTM networks for identifying intrusions [11]. The CICIoMT2024 dataset, introduced in 2024, addresses earlier limitations related to labelled IoMT traffic, thus making it possible to conduct more realistic evaluation. The effectiveness of CNN-LSTM architectures has been observed for identifying network traffic features, especially related to temporal and spatial characteristics.

Implications of these approaches include reducing false positives during real-time monitoring and increasing system robustness. However, challenges still exist with regard to resource constraints of edge devices, which limit the application of complex models in distributed IoMT environments [8], [15].

To highlight key performance metrics, Table I compares top accuracies from selected models:

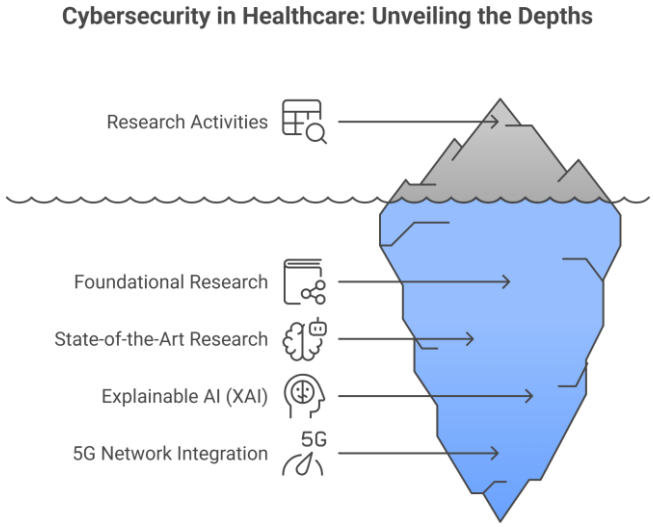
Model	LSTM [11]	CNN-LSTM [10]	Hybrid BERT [14]	Hybrid RL [7]
Accuracy	97%	98%	99.9%	95%

VII. CURRENT EXTENT OF RESEARCH AND IMPLICATIONS FOR CYBERSECURITY

The scope of research activities in the domain of

cybersecurity in healthcare systems is rather vast, incorporating a plethora of research methodologies from basic vulnerability research in networked medical devices to advanced privacy-preserving research using blockchain and federated learning paradigms. While foundational research in this domain has largely focused on creating a comprehensive taxonomy of attacks and their countermeasures, it has created a framework for understanding the attack surface in connected medical environments. Such foundational research has identified critical vulnerabilities in implantable and wearable devices, on-site medical equipment, and communication protocols, along with the need to attenuate these vulnerabilities at the design and regulatory levels.

On the contrary, recent research has continued to advance towards intelligent and adaptive solutions. Current research has continued to place greater emphasis on the integration of explainable artificial intelligence (XAI) to facilitate greater interpretability of intrusion detection systems developed via deep learning techniques to ensure healthcare professionals can better understand and build confidence in decisions made by intrusion detection systems [2], [7]. At the same time, the accelerated adoption of 5G technologies has triggered investigations into ultra-low latency and reliability requirements for critical applications. This is a reflection of the growing awareness of the importance of meeting not only the connectivity requirements of future healthcare environments but also the critical requirements of future 5G wireless infrastructures.



The implications of these research developments on cybersecurity practice are considerable and varied. Improved detection accuracy and lower false positive rates mean that real-time monitoring is more accurate and minimizes disruptions to clinical workflows. Finally, privacy-preserving research developments, especially federated learning, ensure that organizations comply with strict data privacy regulations such as HIPAA and GDPR by not requiring the centralization of patient data during model training [2]. These research developments have created a positive impact on the overall adherence to strict regulations, patient trust in connected health devices, and operational risk reduction for healthcare organizations and medical device manufacturers.

Despite these significant developments, some challenges remain, which affect the progress of the research area. First and foremost, the availability of standardized public datasets, designed specifically for the IoMT environment, remains a significant issue. In fact, the absence of good quality labeled datasets that accurately represent the patterns of IoMT traffic, including heterogeneous device behaviors, 5G, edge computing, and diverse types of attacks, makes it difficult to benchmark, cross-validate, and generally apply the proposed model, as discussed in [6] and [12]. In turn, this makes it difficult to fairly compare the detection systems, thereby impeding the application of the research to the healthcare environment.

Additionally, the resource-constrained nature of devices in distributed IoMT systems continues to pose a challenge in the deployment of computationally intensive deep learning models. In spite of the exploration of various techniques, such as model compression, much work needs to be done to ensure that high-performance intrusion detection systems can be achieved while preserving battery life, processing power, and real-time requirements of devices in distributed IoMT systems, as emphasized in [8] and [15]. Furthermore, while the integration of XAI holds much promise, it is yet to come of age in healthcare applications, where interpretability is a critical requirement.

These open issues, however, emphasize the need to continue to carry out research to bridge the gap between theory and practice in the field of IoMT, which is critical to the deployment of robust, secure, and trustworthy healthcare systems that can withstand the ever-evolving threat landscape in the field of cybersecurity.

VIII. FUTURE RESEARCH

Challenge	**Description**	**Proposed Future Direction**	**Supporting References**
Limited domain-specific datasets	Lack of labelled IoMT traffic for real-world scenarios	Develop and release new public benchmarks (e.g., 5G/IoMT traces)	[2], [12]
Scalability in edge/5G environments	Resource constraints on devices limit complex models	Optimize lightweight hybrid models for edge deployment	[7], [11]
Explainability (XAI)	Black-box DL models reduce clinical trust	Integrate XAI techniques (e.g., SHAP, LIME) in IDS	[4], [14]
Privacy & regulatory compliance	Centralized training risks data exposure	Expand federated learning with differential privacy	[2], [3]

There has been considerable research on the applications of deep learning for the detection of network attacks in healthcare systems. However, there are some research gaps that need to be addressed. For instance, there is a scarcity of domain-related datasets, which makes it difficult for the model to generalize, especially for realistic scenarios related to IoMT, which are characterized by heterogeneous patterns [2], [12]. In addition, scalability in edge and 5G scenarios should be addressed to meet resource and latency-related challenges [7], [11]. The incorporation of Explainable AI is also insufficient, which is an important aspect that should be addressed for healthcare systems, especially for gaining trust and regulatory approvals [4], [14]. Future research should be focused on developing hybrid models that combine federated and reinforcement learning and evaluate them against new benchmarks, including blockchain technology for enhanced security and non-repudiation [3], [15]. There should be additional research on the development of models that can be adapted and change with emerging threats without the need for retraining.

IX. CONCLUSION

Deep learning technology has proven to be an essential tool for identifying and preventing attacks on healthcare networks through its high precision in identifying threats in IoMT environments. The literature review indicates that there are developments in hybrid and federated models that are improving the field. This will help improve the resilience of healthcare systems and enhance the safety of patients despite the rising cyber threats.

REFERENCES

ONE DRIVE: IE3022 Assignment 01 – Sources – IT23545144

[1] A. A. Hady, A. Ghubaish, T. Salman, D. Unal, and R. Jain, "Intrusion detection system for healthcare systems using medical and network data: A comparison study," IEEE Access, vol. 8, pp. 106576–106584, 2020.

[2] A. Rahman et al., "Federated learning-based AI approaches in smart healthcare: Concepts, taxonomies, challenges and open issues," Cluster Comput., vol. 26, pp. 2271–2311, 2023.

[3] T. Yaqoob, H. Abbas, and M. Atiquzzaman, "Security vulnerabilities, attacks, countermeasures, and regulations of networked medical devices—A review," IEEE Commun. Surveys Tuts., vol. 21, no. 4, pp. 3723–3768, 2019.

[4] *M. A. Sahi et al., "Privacy preservation in e-healthcare environments: State of the art and future directions," IEEE Access, vol. 6, pp. 464–478, 2018.*

[5] J. B. Awotunde, S. O. Folorunso, A. L. Imoize, J. O. Odunuga, C.-C. Lee, C.-T. Li, and D.-T. Do, "An ensemble tree-based model for intrusion detection in industrial internet of things networks," Appl. Sci., vol. 13, no. 4, p. 2479, 2023.

[6] M. Zubair et al., "Secure Bluetooth communication in smart healthcare systems: A novel community dataset and intrusion detection system," Sensors, vol. 22, no. 21, p. 8280, 2022.

[7] A. Ghubaish, Z. Yang, and R. Jain, "HDRL-IDS: A hybrid deep reinforcement learning intrusion detection system for enhancing the security of medical applications in 5G networks," in Proc. Int. Conf. Smart Appl. Commun. Netw. (SmartNets), 2024, pp. 1–6.

[8] A. Ghubaish, Z. Yang, A. Erbad, and R. Jain, "LEMDA: A novel feature engineering method for intrusion detection in IoT systems," IEEE Internet Things J., vol. 11, no. 8, pp. 13247–13256, Apr. 2024.

[9] K. Vaisakhkrishnan, G. Ashok, P. Mishra, and T. G. Kumar, "Guarding digital health: Deep learning for attack detection in medical IoT," Procedia Comput. Sci., vol. 235, pp. 2652–2661, 2024.

[10] A. Berguiga, A. Harchay, and A. Massaoudi, "HIDS-IoMT: A deep learning-based intelligent intrusion detection system for the Internet of Medical Things," IEEE Access, vol. 13, pp. 27320–27339, 2025.

[11] J. A. Shaikh et al., "A deep reinforcement learning-based robust intrusion detection system for securing IoMT healthcare networks," Front. Med., vol. 12, 2025, Art. no. 1524286.

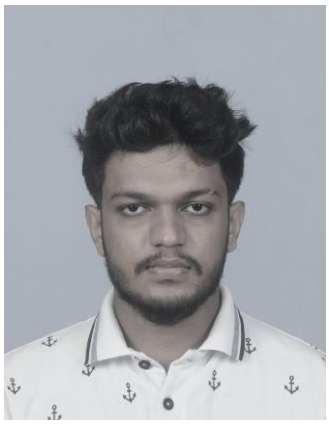
[12] Y. Rbah et al., "Machine learning and deep learning methods for intrusion detection systems in IoMT: A survey," in Proc. 2nd Int. Conf. Innov. Res. Appl. Sci. Eng. Technol. (IRASET), 2022, pp. 1–9.

[13] V. Ravi, T. D. Pham, and M. Alazab, "Deep learning-based network intrusion detection system for Internet of Medical Things," IEEE Internet Things Mag., vol. 6, no. 3, pp. 50–54, Sep. 2023.

[14] A. Alferaidi et al., "A novel hybrid BERT and deep learning model network intrusion detection system for healthcare electronics," IEEE Trans. Consum. Electron., published, doi: 10.1109/TCE.2024.3412199.

[15] A. Alferaidi et al., "Efficient deep learning-based cyber-attack detection for Internet of Medical Things devices," Eng. Proc., vol. 59, no. 1, p. 139, 2023.

ONE DRIVE: IE3022 Assignment 01 – Sources – IT23545144



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He has completed academic and practical projects in security operations, network monitoring, and threat detection, including hands-on experience with Linux systems, network analysis tools, and security monitoring platforms. His technical background includes intrusion detection concepts, vulnerability assessment, packet analysis, and security automation fundamentals. His current research interests focus on deep learning–based intrusion detection, SOC operations, healthcare network security, and protection mechanisms for the Internet of Medical Things (IoMT).

Mr. Peiris is a student member of professional cybersecurity and technology communities. He actively contributes to technical projects and knowledge-sharing platforms related to cybersecurity and network defense. His long-term research interests center on AI-driven security analytics and intelligent threat detection systems for critical infrastructures.